

# A Calculus for Wireless Sensor Networks from Quality Perspective

Xi Wu and Huibiao Zhu

Shanghai Key Laboratory of Trustworthy Computing, East China Normal University

Email: {xiwu, hzbzhu}@sei.ecnu.edu.cn

**Abstract**—Due to deployment constraints and communication modalities, unreliable communications are very common in Wireless Sensor Networks (WSNs). In order to ensure that every node can behave in a reasonable manner even though they are in an unreliable communication, we propose CWQ, a process calculus for formal modeling and reasoning about WSNs and their applications from a quality perspective. We combine local broadcast, one of the most important peculiarities of networks, with quality predicate. Default values are given in the case that ideal behaviors of wireless nodes fail because of the unreliable communication, to increase the service quality offered by the system. Moreover, we also give the operational semantics of CWQ in terms of both a Labeled Transition Semantics and a Reduction Semantics, and prove a correspondence result between them. Finally, some examples, and two real-world case studies Smart Home and Smart Grid are used to illustrate the applicability of our calculus.

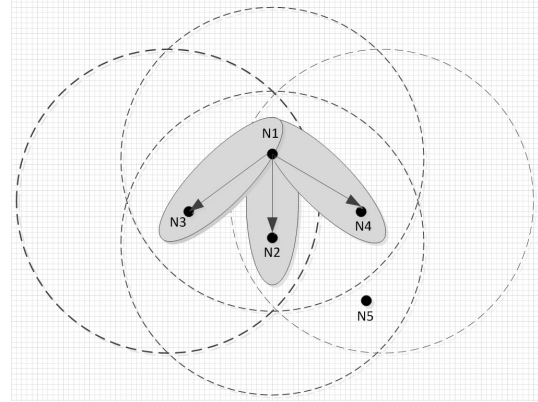


Fig. 1: Local Broadcast Communication Diagram of WSNs

## I. INTRODUCTION

Exploding in popularity, WSNs have been widely used in various fields, including military, ambient intelligence and emergency contingencies. In these kind of large-scale wireless networks, local broadcast is one of the most important peculiarities. Information, which is transmitted in a limited area, can be received by a subset of nodes through wireless links in networks. However, due to the unreliability of links in severe environments, ideal behaviors of the system fail, causing abnormalities and decreasing the quality of service offered by systems, especially for safety-critical systems. How to ensure that all wireless sensor nodes can behave in a reasonable manner (i.e., using approximation values to continue their work) even though they are in an unreliable communication becomes a critical issue. Figure 1 shows the local broadcast communication diagram of WSNs. Only  $N2$ ,  $N3$  and  $N4$ , who are in the transmission area, represented using dashed circle, of sender  $N1$  can receive the broadcast message.

With the rapid development of wireless technology, wireless systems are attracting the attention of many researchers. Up to now, a number of process calculi have been introduced to formalize wireless systems. CCS [9] and CSP [1], [4] are used to describe point-to-point systems. CBS [12], [13] and CBS# [10] provide powerful expressiveness towards one-to-many broadcasting communications in wireless systems. A calculus Timed WSP [6] has been proposed by Si *et al.* for specification and reasoning about WSNs combining together time factor with broadcast communications. In [3], Godskesen has proposed a calculus CMAN for mobile ad hoc networks by treating network topology as part of the network syntax. CMN [7], from an observational theory point of view, has been proposed by Merro to observe the transmission of messages.

Mezzetti *et al.* have proposed CWS [5], [8], which focuses on using semantics to describe interferences among active processes in a wireless system. AWN [2], proposed by Fehnker *et al.*, describes a process algebra for wireless mesh networks by combining local broadcast and data structures. Nonetheless, none of the above takes the service quality offered by the system into account in wireless networks, which is still a challenge.

Nielson *et al.* have proposed the quality calculus [11] and extended it with broadcast communication [14]. However, only process level has been considered without the network level and topological structure. The work may be not well suited for wireless local broadcast since only global broadcast actions (making broadcasted messages reach every node in the network) are allowed. Inspired by [11], in this paper, we develop a calculus for wireless sensor networks from quality perspective (CWQ) and its associated operational semantics for formal modeling and reasoning about WSNs and their applications, in which we combine local broadcast together with quality predicate. In CWQ, topological structure is considered in the network level and different node performances are denoted by processes, i.e., broadcast and receive. In the process level, there is a binder containing quality predicate from [11] as a guard for specifying that the process will continue after the quality predicate being satisfied. The goal of ensuring that every node can behave in a reasonable manner even though they are in an unreliable communication has motivated the creation of CWQ. Here, default values (as the same type as the required ones) are given in the case that ideal behaviors of sensor nodes fail because of the unreliable communication, to increase the service quality offered by the wireless system. Moreover, oper-

Table 1. The Syntax

|                                            |                      |
|--------------------------------------------|----------------------|
| <b>Processes:</b>                          |                      |
| $P, Q ::= nil$                             | termination          |
| $  Act.P$                                  | prefix action        |
| $  P Q$                                    | process parallel     |
| $  \bowtie c.P$                            | change channel       |
| $  P \triangleleft x = y \triangleright Q$ | match                |
| $  A(\tilde{x})$                           | recursion            |
| <b>Act</b> $ ::= b$                        |                      |
| $  t!x$                                    | receive binder       |
| $  t(x)^c$                                 | output               |
| $  !\langle x \rangle^c$                   | broadcast output     |
| <b>b</b> $ ::= t?x$                        |                      |
| $  ?(x)^c$                                 | input                |
| $  \&_q(b_1, \dots, b_n)$                  | broadcast input      |
|                                            | quality predicate    |
| <b>Networks:</b>                           |                      |
| $M, N ::= 0$                               | empty network        |
| $  n[P]_{l,r}^{c:C}$                       | node                 |
| $  M  N$                                   | parallel composition |
| <b>Function:</b>                           |                      |
| $D(l_1, l_2)$                              | distance function    |

ational semantics is given in terms of both Labeled Transition Semantics (LTS) and Reduction Semantics (RS). Even though LTS can not only describe the internal actions of the system, but also characterize the interactions between the system and its surrounding environment, however, some LTS rules are very hard to grasp. Thus, in this paper, we also define RS for CWQ, which is easier to understand. Inspired of Davide Sangiorgi's work [5], we prove correspondence results between LTS and RS to show the correctness of the operational semantics using Harmony Theorem instead of bisimulation. Finally, we model and analyze two real-world scenarios, smart home and smart grid, to demonstrate the applicability of our calculus. The main contributions of this paper are listed as follows:

- **Process calculus:** a calculus for WSNs combining local broadcast together with service quality, considering topological structure is presented.
- **Operational semantics:** both LTS and RS with correspondence results between them are given.
- **Application:** two of the most promising real-world scenarios, smart home and smart grid, to illustrate the applicability of CWQ are used.

The rest of the paper is organized as follows. Section 2 provides the syntax of CWQ. Section 3 and section 4 present the operational semantics in terms of both LTS and RS respectively. A correspondence result between them proved using harmony theorem is also given in section 4. Section 5 comprises real-world case studies about smart home and smart grid, showing the applicability of CWQ. A conclusion summarizing this paper and foreseeable future work is presented in section 6.

## II. CWQ CALCULUS

We define the notion of CWQ in two parts: process part which is used to describe the possible status of a node, and network part for representing the communication in the entire system. The syntax is shown in Table 1. We employ  $P, Q$  to range over the set of all processes, and  $M, N$  the set of all nodes. We also use set  $\mathcal{I}_n$  to denote node identities (or names), where  $n_1, n_2, \dots$  range over it.

- Note that, at some time, some variables in the binder do not get values because the corresponding inputs have not occurred, thus there is one special notation *none* to stand for the absence of data.

The syntax of our calculus is shown by the Backus-Naur form. *nil* stands for the skip process.  $t!x$  refers to outputting a value  $x$  on channel  $t$  while  $t?x$  represents inputting a value on channel  $t$ . Note that, in a local process, an input action can synchronize with an output one via some channel.  $!\langle x \rangle^c$  denotes broadcasting the value  $x$ , while the corresponding reception of a broadcasted value is represented by  $?(x)^c$ . Here,  $c$  is a broadcasting channel.  $P|Q$  means that two processes run in parallel and synchronize via shared channels;  $P \triangleleft x = y \triangleright Q$  behaves as  $P$  if  $x$  and  $y$  match, and otherwise as  $Q$ ;  $A(\tilde{x})$  denotes a process with the (possibly recursive) definition of  $A(\tilde{x}) =_{df} P$ , where  $A$  is a process constant and  $\tilde{x}$  contains all variables that appear free in  $P$ . The operator  $\bowtie$  refers to changing channel, e.g., a node currently using broadcasting channel  $c_1$  can change the channel from  $c_1$  into  $c_2$  via  $\bowtie c_2$ . Moreover, in order to decide whether a node is in or out of the transmission cell of another node, we define a function  $D$  which takes two locations as its parameters and returns the distance between these two locations.

In CWQ, we inherit the form  $\&_q(b_1, \dots, b_n)$  in the receive binder  $b$  from [11], which is used as a guard for specifying that the process will continue after the quality predicate being satisfied.  $n$  is the total number of inputs and  $q$  is a quality predicate to be satisfied, indicating when sufficient inputs have been received to continue. Here,  $q \in \{\forall, \exists, m/n\}$  and we give the details below.

- $\forall$ : all inputs are required, e.g.,  $\&_{\forall}(t_1?x_1, t_2?x_2, ?(x_3)^c)$  requires three sufficient inputs and it has the same effect as  $\&_q(t_1?x_1, t_2?x_2, ?(x_3)^c)$  if  $q(r_1, r_2, r_3)$  amounts to  $r_1 \wedge r_2 \wedge r_3$ .
- $\exists$ : at least one of the inputs is required, e.g.,  $\&_{\exists}(t_1?x_1, t_2?x_2, ?(x_3)^c)$  requires one sufficient input to continue and it has the same effect as  $\&_q(t_1?x_1, t_2?x_2, ?(x_3)^c)$  if  $q(r_1, r_2, r_3)$  amounts to  $r_1 \vee r_2 \vee r_3$ .
- $m/n$ :  $m$  sufficient inputs of all  $n$  inputs are required to be accepted, e.g.,  $\&_{m/3}(t_1?x_1, t_2?x_2, ?(x_3)^c)$  requires at least two sufficient inputs either from the point-to-point channel or from the broadcast channel.

Besides, we also allow the nested binders such as  $\&_{\exists}(\&_{\forall}(t_1?x_1, t_2?x_2), ?(x_3))$ , which describes that input must be received either over the broadcast channel or both over channels  $t_1$  and  $t_2$ .

Networks are collections of nodes running in parallel. Hence at the network level, we have the parallel composition of networks  $M||N^1$ , each of which is inductively defined by parallel composition of an empty network  $0$  and a wireless node written as  $n[P]_{l,r}^{c:C}$ . Each node is assigned a unique identity  $n$ , and runs some process  $P$  with physical location  $l$ , dissemination radius  $r$  and currently broadcast channel  $c$ . Each node has an available channel set  $C$ , which means that one node has ability to broadcast and receive messages from

<sup>1</sup>Multiple composition of networks can be written as  $||_{i \in \mathcal{I}} M_i$ .

Table 2. Structural Congruence

|                                                                             |                                    |
|-----------------------------------------------------------------------------|------------------------------------|
| $n[P Q]_{l,r}^{c,C} \equiv n[Q P]_{l,r}^{c,C}$                              | Communicative Law for Process      |
| $N  0 \equiv N$                                                             | Identical Law in the Empty Network |
| $N  N' \equiv N'    N$                                                      | Communicative Law                  |
| $N  (N'    N'') \equiv (N    N')    N''$                                    | Associative Law                    |
| $n[A(u)]_{l,r}^c \equiv n[P\{u/x\}]_{l,r}^c$ , if $A(x) \stackrel{df}{=} P$ | Equivalence Law of Recursion       |

some different channels, but it can only use one channel at one time.

### III. LABELED TRANSITION SEMANTICS

We first expect a structural congruence as an auxiliary relation, represented by  $\equiv$ . It allows us to do the operations on network structures to put participants with their potential interactions into parallel positions. We just give the minimal set of structural congruence in Table 2 and the meanings of all the formulas are very intuitive, thus we do not give the expressions.

#### A. Labeled Transitions for Processes

In our paper, LTS is divided into two parts: labeled transitions for processes and labeled transitions for network. The form of labeled transition rules for processes is like  $P \xrightarrow{\alpha} P'$ , which can be found in Table 3. The syntax of the signal  $\alpha$  is defined as follows:

$$\alpha := !\langle v \rangle^c \mid ?(x)^c$$

$!\langle v \rangle^c$  stands for broadcasting message  $v$  via channel  $c$ . The action  $?(x)^c$  represents receiving a broadcast message through channel  $c$ , then the value of the message will be assigned to variable  $x$ . Some auxiliary relations are given first:

$$b \check{Q}_t \theta \text{ and } b \check{Q}_f \theta$$

to describe whether the required inputs in binder  $b$  have already been satisfied ( $\check{Q}_t$ ) or not ( $\check{Q}_f$ ). If all the required inputs are satisfied, a substitution  $\theta$  is constructed to replace all the variables with the receiving values. More details are shown below:

$$\begin{cases} ?(x)^c \check{Q}_t [y/x], & t?x \check{Q}_t [y/x]; \\ ?(x)^c \check{Q}_f [none/x], & t?x \check{Q}_f [none/x]. \end{cases}$$

These rules denote the details of substitution  $\theta$ . At some time, if a receiving happens, the value of variable  $x$  can be replaced by some present value  $y$ , otherwise, the value will be replaced by special notation *none*.

$$!\langle v \rangle^c \vdash b \rightarrow b' \text{ and } t!v \vdash b \rightarrow b'$$

specify that the binder  $b$  changes into  $b'$  after receiving broadcast output  $!\langle v \rangle^c$  or output  $t!v$  respectively.

$$c \in C \text{ and } c \notin C$$

are used to check whether channel  $c$  is in some node's available channel set  $C$  or not. If it is, then the node can receive messages via  $c$ .

Rule Bro\_Send defines process  $!\langle v \rangle^c.P$  is evolved into  $P$  after broadcasting message  $v$ . Rule Bro\_Receive stands for receiving. Rule Qli\_Bro1 and Qli\_Bro2 are used for describing the synchronization of the broadcast actions with quality. The former defines that after the binder  $b$  receiving a value from broadcast output, the required inputs in  $b$  still cannot be

satisfied, thus more inputs are required and  $b$  changes into  $b'$ . While the later describes no more inputs are needed after binder  $b$  receiving the broadcast message and a substitution  $\theta$  is constructed to replace the variables in  $P$ . Rule Case1 and Case2 stand for matching. If  $x$  equals to some data  $y$ , then the node continues to execute the process  $P_1$ , otherwise the node executes the process  $P_2$ .

Similar with Qli\_Bro1 and Qli\_Bro2, rule Quality\_Syn1 and Quality\_Syn2 are used for describing the synchronization of outputting with quality instead of broadcast outputting. Rule Quality\_Rec1 and Quality\_Rec2 describe a complex situation that any binder  $b_i$  receiving no matter a broadcast output message or an output message can be evolved into a new binder  $b'_i$ . The last rule Satisfy shows how to check whether the quality predicate in the form  $\&_q(b_1, \dots, b_i, \dots, b_n)$  is satisfied or not. At some time, each binder in the form  $\&_q(b_1, \dots, b_i, \dots, b_n)$  can be checked using the auxiliary relations  $\check{Q}_t$  and  $\check{Q}_f$  we give before.

#### B. Labeled Transitions for Network

The form of the transition rules is  $N \xrightarrow{\mu} N'$ . The symbol  $\xrightarrow{\mu}$  includes  $\xrightarrow{c!!\langle v \rangle[l,r]}$ ,  $\xrightarrow{c??(v)@l}$  and  $\xrightarrow{\tau}$ . Here,  $\xrightarrow{c!!\langle v \rangle[l,r]}$  represents that a node located in  $l$  with the transmission radius  $r$ , broadcasts a message via channel  $c$ .  $\xrightarrow{c??(v)@l}$  means that a node receives a broadcast message which is sent at position  $l$ . Moreover,  $\xrightarrow{\tau}$  stands for internal event. First, we define one rule for the recursion operator Net\_Rec as follows:

$$[Net\_Rec] \quad \frac{n[P\{u/x\}]_{l,r}^{c,C} \xrightarrow{\mu} n[P']_{l,r}^{c,C} \quad A(x) \stackrel{df}{=} P}{n[A(u)]_{l,r}^c \xrightarrow{\mu} n[P']_{l,r}^{c,C}}$$

Labeled transition rules for network can be found in Table 4. Empty network does nothing with any event, which is shown in rule Net\_Null1 and Net\_Null2. Rule Net\_Bro denotes that a node, located in the position  $l$ , wants to broadcast a message  $v$  via the broadcast channel  $c$  with the radius of  $r$ . After broadcasting, the executing process  $P$  evolves into  $P'$ . Rule Net\_Rcv1 is used to describe that a node, in the transmission cell of the sending node located in  $l$ , receives the message  $v$  through broadcast channel  $c'$  and executes the process  $P'$ . If the node is not in the transmission cell of the sending node or the broadcasting channel  $c'$  is not in its available channel set, the node will be unchanged, which is shown in rule Net\_Rcv2.

Rule Net\_Par1 and Net\_Par2 specify the parallel of the networks as  $N_2$  can receive the message or not. Rule Net\_Rcv3 describes that both of the networks  $N_1$  and  $N_2$  can do the receiving action and they still wait for receiving messages after the parallel. Rule Net\_Int describes the internal actions inside one node. Rule Net\_Par3 is a combination rule. Because the internal events do not impact on the external environment, thus the network executing the internal event can parallel with other networks unconditionally. Rule Net\_Switch represents changing channel from  $c$  to  $c'$ .

#### C. Example for Labeled Transition Semantics

An example is given here including one broadcast node  $n_1[!\langle v_0 \rangle^{c_1}.P_1|Q_1]_{l_1,r_1}^{c_1:\{c_1\}}$ , and three receiving nodes  $n_2, n_3$  and  $n_4$ , in which  $n_2$  and  $n_4$  are in the transmission cell of  $n_1$ , but  $n_3$  is out of the transmission range of the sending node. The structure of the example network is listed below:

**Table 3. Labeled Transitions for Processes**

|                                                                                                                                                                                                                                                                                                             |                                                                                                                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| $[Bro\_Send] \quad !\langle v \rangle^c.P \xrightarrow{!(v)^c} P$                                                                                                                                                                                                                                           | $[Bro\_Receive] \quad ?(x)^c.P \xrightarrow{?(v)^c} P[v/x]$                                                                                    |
| $[Qli\_Bro1] \quad \frac{!(v)^c \vdash b \rightarrow b' \quad b' \delta_f \theta}{b.P \xrightarrow{?(x)^c} b'.P}$                                                                                                                                                                                           | $[Qli\_Bro2] \quad \frac{!(v)^c \vdash b \rightarrow b' \quad b' \delta_f \theta}{b.P \xrightarrow{?(x)^c} P\theta}$                           |
| $[Case1] \quad \frac{x=y}{P_1 \triangleleft x=y \triangleright P_2 \rightarrow P_1}$                                                                                                                                                                                                                        | $[Case2] \quad \frac{x \neq y}{P_1 \triangleleft x=y \triangleright P_2 \rightarrow P_2}$                                                      |
| $[Quality\_Syn1] \quad \frac{t_1!v_1 \vdash b \rightarrow b' \quad b' \delta_f \theta}{t_1!v_1.P_1 \mid b.P_2 \rightarrow P_1 \mid b'.P_2}$                                                                                                                                                                 | $[Quality\_Syn2] \quad \frac{t_1!v_1 \vdash b \rightarrow b' \quad b' \delta_f \theta}{t_1!v_1.P_1 \mid b.P_2 \rightarrow P_1 \mid P_2\theta}$ |
| $[Quality\_Rec1] \quad \frac{!(v)^c \vdash b_i \rightarrow b'_i}{!(v)^c \vdash \&q(b_1, \dots, b_i, \dots, b_n) \rightarrow \&q(b_1, \dots, b'_i, \dots, b_n)}$                                                                                                                                             |                                                                                                                                                |
| $[Quality\_Rec2] \quad \frac{t_1!v_1 \vdash b_i \rightarrow b'_i}{t_1!v_1 \vdash \&q(b_1, \dots, b_i, \dots, b_n) \rightarrow \&q(b_1, \dots, b'_i, \dots, b_n)}$                                                                                                                                           |                                                                                                                                                |
| $[Satisfy] \quad \frac{b_1 \delta_{\sigma_1} \theta_1 \dots b_i \delta_{\sigma_i} \theta_i \dots b_n \delta_{\sigma_n} \theta_n}{\&q(b_1, \dots, b_i, \dots, b_n) \delta_{\sigma} \theta_1 \dots \theta_i \dots \theta_n} \quad \text{where } \sigma = [\{q\}](\sigma_1, \dots, \sigma_i, \dots, \sigma_n)$ |                                                                                                                                                |

**Table 4. Labeled Transitions for Network**

|                                                                                                                                                                              |                                                                                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $[Net\_Bro] \quad \frac{P \xrightarrow{!(v)^c} P'}{n[P Q]_{l,r}^{c:C} \xrightarrow{c!!(v)[l,r]} n[P' Q]_{l,r}^{c:C}}$                                                        | $[Net\_Rcv1] \quad \frac{P \xrightarrow{?(v)^c} P' \quad D(l, l') \leq r' \wedge c' \in C}{n[P Q]_{l,r}^{c:C} \xrightarrow{c'??(v) \otimes l'} n[P' Q]_{l,r}^{c':C}}$                  |
| $[Net\_Rcv2] \quad \frac{D(l, l') > r' \vee c' \notin C}{n[P Q]_{l,r}^{c:C} \xrightarrow{c'??(v) \otimes l'} n[P Q]_{l,r}^{c':C}}$                                           | $[Net\_Rcv3] \quad \frac{N_1 \xrightarrow{c??(v) \otimes l} N'_1 \quad N_2 \xrightarrow{c??(v) \otimes l} N'_2}{N_1 \parallel N_2 \xrightarrow{c??(v) \otimes l} N'_1 \parallel N'_2}$ |
| $[Net\_Par1] \quad \frac{N_1 \xrightarrow{c!!(v)[l,r]} N'_1 \quad N_2 \xrightarrow{c??(v) \otimes l} N'_2}{N_1 \parallel N_2 \xrightarrow{c!!(v)[l,r]} N'_1 \parallel N'_2}$ | $[Net\_Par2] \quad \frac{N_1 \xrightarrow{c!!(v)[l,r]} N'_1 \quad N_2 \xrightarrow{c??(v) \otimes l} N'_2}{N_1 \parallel N_2 \xrightarrow{c!!(v)[l,r]} N'_1 \parallel N'_2}$           |
| $[Net\_Null1] \quad 0 \xrightarrow{c??(v) \otimes l} 0$                                                                                                                      | $[Net\_Null2] \quad 0 \xrightarrow{\tau} 0$                                                                                                                                            |
| $[Net\_Int] \quad \frac{P \xrightarrow{\tau} P'}{n[P]_{l,r}^{c:C} \xrightarrow{\tau} n[P']_{l,r}^{c:C}}$                                                                     |                                                                                                                                                                                        |
| $[Net\_Switch] \quad n[\&\exists c'.P Q]_{l,r}^{c:\{c,c'\}} \xrightarrow{\tau} n[P Q]_{l,r}^{c':\{c,c'\}}$                                                                   | $[Net\_Par3] \quad \frac{N_1 \xrightarrow{\tau} N'_1}{N_1 \parallel N_2 \xrightarrow{\tau} N'_1 \parallel N_2}$                                                                        |
|                                                                                                                                                                              | $N_2 \parallel N_1 \xrightarrow{\tau} N_2 \parallel N'_1$                                                                                                                              |

$$\begin{aligned}
N &\stackrel{df}{=} n_1[!(v_0)^{c_1}.P_1|Q_1]_{l_1, r_1}^{c_1:\{c_1\}} \parallel \\
&\quad n_2[\&\forall?(x_0)^{c_1}, t_1?x_1).P_2|t_1!v_1.Q_2]_{l_2, r_2}^{c_1:\{c_1, c_2\}} \parallel \\
&\quad n_3[\&\exists(t_2?x_2, t_3?x_3).P_3|t_2!v_2.Q_3]_{l_3, r_3}^{c_1:\{c_1, c_3\}} \parallel \\
&\quad n_4[\&\exists?(x_4)^{c_1}, t_4?x_4).P_4|Q_4]_{l_4, r_4}^{c_1:\{c_1, c_4\}}
\end{aligned}$$

Where  $D(l_1, l_2) \leq r_1 \wedge c_1 \in \{c_1, c_2\}$ ,  $D(l_1, l_3) > r_1$ ,  $D(l_1, l_4) \leq r_1 \wedge c_1 \in \{c_1, c_4\}$ . Then there is one possible labeled transition situation below:

$$\begin{aligned}
N &\xrightarrow{c!!(v_0)[l_1, r_1]} n_1[P_1|Q_1]_{l_1, r_1}^{c_1:\{c_1\}} \\
&\quad \parallel n_2[\&\forall([v_0/x_0], [v_1/x_1]).P_2|t_1!v_1.Q_2]_{l_2, r_2}^{c_1:\{c_1, c_2\}} \\
&\quad \parallel n_3[\&\exists(t_2?x_2, t_3?x_3).P_3|t_2!v_2.Q_3]_{l_3, r_3}^{c_1:\{c_1, c_3\}} \\
&\quad \parallel n_4[\&\exists([v_0/x_4], [t_4?x_4]).P_4|Q_4]_{l_4, r_4}^{c_1:\{c_1, c_4\}} \\
&\stackrel{df}{=} N_1
\end{aligned}$$

$$\begin{aligned}
N_1 &\xrightarrow{\tau} n_1[P_1|Q_1]_{l_1, r_1}^{c_1:\{c_1\}} \parallel \\
&\quad n_2[\&\forall([v_0/x_0], [v_1/x_1]).P_2|Q_2]_{l_2, r_2}^{c_1:\{c_1, c_2\}} \parallel \\
&\quad n_3[\&\exists([v_2/x_2], [t_3?none/x_3]).P_3|Q_3]_{l_3, r_3}^{c_1:\{c_1, c_3\}} \parallel \\
&\quad n_4[\&\exists([v_0/x_4], [t_4?none/x_4]).P_4|Q_4]_{l_4, r_4}^{c_1:\{c_1, c_4\}} \stackrel{df}{=} N_2 \\
N_2 &\xrightarrow{\tau} n_1[P_1|Q_1]_{l_1, r_1}^{c_1:\{c_1\}} \parallel n_2[P_2|Q_2]_{l_2, r_2}^{c_1:\{c_1, c_2\}} \parallel \\
&\quad n_3[P_3|Q_3]_{l_3, r_3}^{c_1:\{c_1, c_3\}} \parallel n_4[P_4|Q_4]_{l_4, r_4}^{c_1:\{c_1, c_4\}}
\end{aligned}$$

The quality predicate in node  $n_2$  is  $\forall$  and both two inputs are satisfied, thus the process evolves into  $P_2$ . Nodes  $n_3$  and  $n_4$  have the predicate  $\exists$ , thus the executing processes in them evolve into  $P_3$  and  $P_4$  respectively.

## IV. REDUCTION SEMANTICS

Even though LTS is very popular, some LTS rules are very hard to grasp. Thus, in this paper, we also define RS for CWQ, which is easier to understand. Inspired of Davide Sangiorgi's work [5], we also prove correspondence results between LTS and RS to show the correctness of the operational semantics using Harmony Theorem instead of bisimulation.

### A. Reduction Semantics

Here, a reduction is either a broadcast event or an internal event. Broadcast event is represented by  $\leadsto_{l,r}^c$ , which means that the node located in position  $l$  wants to broadcast message via the current broadcast channel  $c$  within the transmission radius  $r$ . In Table 5, Rule RS-Bcast wants to broadcast message  $v$  via the broadcast channel  $c$  to all nodes. Nodes who are located inside the transmission cell  $\forall i \in I, D(l, l_i) \leq r$  and monitoring the broadcast channel  $c$  will receive the broadcast message. While the other nodes who are out of the transmission cell  $\forall j \in J, D(l, l_j) > r$  or  $c$  is not in their available channel set  $C_j$ , will not receive the broadcast message. Here,  $I$  and  $J$  are sets of nodes. After broadcasting, the sending node will be changed into the form of  $n[P|Q]_{l,r}^{c:C}$  and the receiving nodes in set  $I$  will be changed into the form of  $\&q_i(b_1, b_2, \dots, b'_k, \dots, b_n).P_i|Q_i$ , while nodes in set  $J$  will be remained unchanged.

Rule RS-Null shows that empty network does nothing to the broadcast event. The closure rule RS-Cont focuses

Table 5. Reduction Semantics

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| $[RS - Bcast] \frac{\forall i \in I. D(l, l_i) \leq r \wedge c \in C_i \quad \forall j \in J. D(l, l_j) > r \vee c \notin C_j \quad !\langle v \rangle^c \vdash b_k \rightarrow b'_k}{n[\langle v \rangle^c.P Q]_{l,r}^{c:C} \parallel \prod_{i \in I, n \in N, k \in N, k \leq n} n_i[\&q_i(b_1, b_2, \dots, b_k, \dots, b_n).P_i Q_i]_{l_i, r_i}^{c_i:C_i} \parallel \prod_{j \in J, m \in N, h \in N, h \leq m} n_j[\&q_j(b_1, b_2, \dots, b_h, \dots, b_m).P_j Q_j]_{l_j, r_j}^{c_j:C_j} \sim_{l,r}^{c:C} n[P Q]_{l,r}^{c:C} \parallel \prod_{i \in I, n \in N, k \in N, k \leq n} n_i[\&q_i(b_1, b_2, \dots, b'_k, \dots, b_n).P_i Q_i]_{l_i, r_i}^{c_i:C_i} \parallel \prod_{j \in J, m \in N, h \in N, h \leq m} n_j[\&q_j(b_1, b_2, \dots, b_h, \dots, b_m).P_j Q_j]_{l_j, r_j}^{c_j:C_j}}$ |  |
| $[RS - Cont] \frac{N \sim_{l,r}^{c:C} N' \quad (l, r, c) \nparallel N''}{N \parallel N'' \sim_{l,r}^{c:C} N' \parallel N''} \quad [RS - Null] \quad 0 \sim_{l,r}^c 0 \quad [RS - Congr] \quad \frac{M \equiv N \quad N \sim_{l,r}^{c:C} N' \quad N' \equiv M'}{M \sim_{l,r}^{c:C} M'}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
| $[RS - Inter] \frac{\forall i \in I \quad t!v \vdash b_k \rightarrow b'_k}{\parallel \prod_{i \in I, n \in N, k \in N, k \leq n} n_i[\&q_i(b_1, b_2, \dots, b_k, \dots, b_n).P_i t_k!v_k.Q_i]_{l_i, r_i}^{c_i:C_i} \sim \parallel \prod_{i \in I, n \in N, k \in N, k \leq n} n_i[\&q_i(b_1, b_2, \dots, b'_k, \dots, b_n).P_i Q_i]_{l_i, r_i}^{c_i:C_i}}$                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |
| $[RS - Choice - True] \frac{x=y}{n[P_1 \triangleleft x=y \triangleright P_2 Q]_{l,r}^{c:C} \sim n[P_1 Q]_{l,r}^{c:C}} \quad [RS - Choice - False] \frac{x \neq y}{n[P_1 \triangleleft x=y \triangleright P_2 Q]_{l,r}^{c:C} \sim n[P_2 Q]_{l,r}^{c:C}}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |
| $[RS - ST] \frac{b' \hat{Q}_{t,f} \theta}{n[b'.P Q]_{l,r}^{c:C} \sim n[P\theta Q]_{l,r}^{c:C}} \quad [RS - SF] \frac{b' \hat{Q}_{f,\theta}}{n[b'.P Q]_{l,r}^{c:C} \sim n[b'.P Q]_{l,r}^{c:C}} \quad [RS - Null - Int] \quad 0 \sim 0$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |
| $[RS - Switch] \quad n[\bowtie c'.P Q]_{l,r}^{c:\{c,c'\}} \sim n[P Q]_{l,r}^{c:\{c,c'\}} \quad [RS - Cont - Int] \quad \frac{N \sim_{l,r}^{c:C} N'}{N \parallel N'' \sim_{l,r}^{c:C} N' \parallel N''}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |

on the parallel of the network. A condition  $(l, r, c) \nparallel N''$  denotes that no nodes in network  $N''$  can receive the broadcast message from the node in  $l$  via channel  $c$ , thus  $N''$  can be unconditionally paralleled with the network  $N$  without any effect. Rule RS-Congr describes the network will be evolved into a new structural congruence network after broadcast event.

$\sim$  stands for the internal event, happened inside the node, which cannot be seen from the network perspective. Rule RS-Inter is used for the communication between processes inside any node in set  $I$ , e.g., output and input. Rule RS-Choice-True and rule RS-Choice-False describe the internal matching structure. Rule RS-ST and rule RS-SF are used to check whether the quality predicate is satisfied or not, and the symbols  $\hat{Q}_{\{t,f\}}$  have already been introduced in last section. Node can change its current broadcast channel using rule RS-Switch and rule RS-Null-Int represents that empty network has no effect to internal event. Finally, rule RS-Cont-Int is a closure rule denoting that any network can parallel with the network executing an internal event.

### B. Example for Reduction Semantics

An example is given to illustrate how to use CWQ RS rules. The sending node  $n_1$ , located in  $l_1$  with the transmission radius  $r_1$ , wants to broadcast a message to all nodes via its current broadcast channel  $c_1$ . Node  $n_2$ , inside the transmission cell of  $n_1$ , will receive the broadcast message, but  $n_3$ , out of the transmission cell of node  $n_1$ , cannot hear the message clearly. The detailed structure of the network is as follows:

$$N \stackrel{df}{=} n_1[\langle v_0 \rangle^{c_1}.P_1|Q_1]_{l_1, r_1}^{c_1:\{c_1\}} \parallel n_2[\&\forall?(x_0)^{c_1}, t_1?x_1).P_2[t_1!v_1.Q_2]_{l_2, r_2}^{c_1:\{c_1, c_2\}} \parallel n_3[\&\exists(t_2?x_2, t_3?x_3).P_3|Q_3]_{l_3, r_3}^{c_1:\{c_1, c_3\}}$$

Where  $D(l_1, l_2) \leq r_1 \wedge c_1 \in \{c_1, c_2\}$  and  $D(l_1, l_3) > r_1 \wedge c_1 \in \{c_1, c_3\}$ . Then there is one possible reduction situation below:

$$N \sim_{l_1, r_1}^{c_1} n_1[P_1|Q_1]_{l_1, r_1}^{c_1:\{c_1\}} \parallel n_2[\&\forall(\langle v_0/x_0 \rangle, t_1?x_1).P_2[t_1!v_1.Q_2]_{l_2, r_2}^{c_1:\{c_1, c_2\}} \parallel n_3[\&\exists(t_2?x_2, t_3?x_3).P_3|Q_3]_{l_3, r_3}^{c_1:\{c_1, c_3\}} \stackrel{df}{=} N_1$$

$$N_1 \sim n_1[P_1|Q_1]_{l_1, r_1}^{c_1:\{c_1\}} \parallel$$

$$n_2[\&\forall(\langle v_0/x_0 \rangle, [v_1/x_1]).P_2|Q_2]_{l_2, r_2}^{c_1:\{c_1, c_2\}} \parallel n_3[\&\exists(t_2?[none/x_2], t_3?[none/x_3]).P_3|Q_3]_{l_3, r_3}^{c_1:\{c_1, c_3\}} \stackrel{df}{=} N_2$$

$$N_2 \sim n_1[P_1|Q_1]_{l_1, r_1}^{c_1:\{c_1\}} \parallel n_2[P_2|Q_2]_{l_2, r_2}^{c_1:\{c_1, c_2\}} \parallel n_3[\&\exists(t_2?x_2, t_3?x_3).P_3|Q_3]_{l_3, r_3}^{c_1:\{c_1, c_3\}}$$

We will prove the correspondence between LTS and RS by harmony theorem.

### C. Harmony Theorem

The operational semantics of CWQ is given in terms of both LTS and RS in section 3 and section 4 respectively. In this subsection, we prove a correspondence result between them.

First, it gives a proof that the equivalence in the structural congruence is also suitable for LTS. Secondly, each rule in RS can find a corresponding rule in LTS. Finally, it proves that each rule in LTS can also find a corresponding rule in RS.

**Theorem 1 (Harmony Theorem)** *Let  $M$  and  $N$  be networks.*

- 1) *If  $M \xrightarrow{\sigma} M'$  and  $M \equiv N$  then there exists  $N'$  such that  $N \xrightarrow{\sigma} N' \equiv M'$*
- 2) *(1) If  $M \sim_{l,r}^c M'$  then  $M \xrightarrow{c!!\langle v \rangle[l,r]} M_1 \equiv M'$*   
*(2) If  $M \sim M'$  then  $M \xrightarrow{\tau} M_1 \equiv M'$*
- 3) *The converse also holds.*

**Proof:** We give the proofs for these three points respectively as follows:

1) Equivalence relations in structural congruence is shown in Table 2 and we will prove that all these equivalence relations are held in LTS as well. Rule Net\_Null1 and rule Net\_Null2 let the Identical Law in the Empty Network hold, which show that the empty network will never affect and change the environment. Communicative Law is held because in all LTS rules, rule Net\_Par1 and Net\_Par2 are symmetrical, and the other rules e.g., Net\_Rcv3 and Net\_Int, are reflexive. Moreover, each operator satisfies Associative Law and the network structure has always been able to be maintained, which makes the Associative Law be guaranteed by the

whole network. Finally, rule Net\_Rec makes sure that the possible transition rules are same for the network no matter there exists the recursion operator or not, which makes the Equivalence Law of Recursion hold.

2) The second part of the theorem is proved by induction of these two derivations  $M \rightsquigarrow_{l,r}^c M'$  and  $M \rightsquigarrow M'$ .

(1) First, we consider that the derivation  $M \rightsquigarrow_{l,r}^c M'$  is generated by an application of the reduction rule RS-Bcast, which is,

$$\begin{aligned} & n[!(v)^c.P|Q]_{l,r}^{c:C} \\ & \parallel \prod_{i \in I, n \in \mathbb{N}, k \in \mathbb{N}, k \leq n} n_i[\&_{q_i}(b_1, b_2, \dots, b_k, \dots, b_n).P_i|Q_i]_{l_i, r_i}^{c:C_i} \\ & \parallel \prod_{j \in J, m \in \mathbb{N}, h \in \mathbb{N}, h \leq m} n_j[\&_{q_j}(b_1, b_2, \dots, b_h, \dots, b_m).P_j|Q_j]_{l_j, r_j}^{c:C_j} \\ \rightsquigarrow_{l,r}^c & n[P|Q]_{l,r}^{c:C} \\ & \parallel \prod_{i \in I, n \in \mathbb{N}, k \in \mathbb{N}, k \leq n} n_i[\&_{q_i}(b_1, b_2, \dots, b'_k, \dots, b_n).P_i|Q_i]_{l_i, r_i}^{c:C_i} \\ & \parallel \prod_{j \in J, m \in \mathbb{N}, h \in \mathbb{N}, h \leq m} n_j[\&_{q_j}(b_1, b_2, \dots, b_h, \dots, b_m).P_j|Q_j]_{l_j, r_j}^{c:C_j} \end{aligned}$$

such that  $D(l, l_i) \leq r \wedge c \in C_i$  and  $D(l, l_j) > r$ , for all  $i \in I$  and  $j \in J$ . Then, these two derivations below are valid.

$$\begin{aligned} & \frac{\frac{P \xrightarrow{!(v)^c} P'}{n[P|Q]_{l,r}^{c:C} \xrightarrow{c!!(v)[l,r]} n[P'|Q]_{l,r}^{c:C}} \quad \frac{P_1 \xrightarrow{?(v)^c} P'_1 \quad D(l_1, l) \leq r \wedge c \in C}{n_1[P_1|Q_1]_{l_1, r_1}^{c:C} \xrightarrow{c!!(v)[l,r]} n_1[P'_1|Q_1]_{l_1, r_1}^{c:C}} \quad \frac{P_2 \xrightarrow{?(v)^c} P'_2 \quad D(l_2, l) > r \vee c \notin C'}{n_2[P_2|Q_2]_{l_2, r_2}^{c:C'} \xrightarrow{c!!(v)[l,r]} n_2[P'_2|Q_2]_{l_2, r_2}^{c:C'}}}{\frac{n[P|Q]_{l,r}^{c:C} \parallel n_1[P_1|Q_1]_{l_1, r_1}^{c:C} \parallel n_2[P_2|Q_2]_{l_2, r_2}^{c:C'} \xrightarrow{c!!(v)[l,r]} n[P|Q]_{l,r}^{c:C} \parallel n_1[P'_1|Q_1]_{l_1, r_1}^{c:C} \parallel n_2[P'_2|Q_2]_{l_2, r_2}^{c:C'}}} \end{aligned}$$

By applying  $|I|$  times labeled transition rule Net\_Par1 and  $|J|$  times rule Net\_Rcv3, we can get:

$$\begin{aligned} & n[P|Q]_{l,r}^{c:C} \parallel \prod_{i \in I} n_i[P_i|Q_i]_{l_i, r_i}^{c:C_i} \parallel \prod_{j \in J} n_j[P_j|Q_j]_{l_j, r_j}^{c:C_j} \xrightarrow{c!!(v)[l,r]} \\ & n[P'|Q]_{l,r}^{c:C} \parallel \prod_{i \in I} n_i[P'_i|Q_i]_{l_i, r_i}^{c:C_i} \parallel \prod_{j \in J} n_j[P'_j|Q_j]_{l_j, r_j}^{c:C_j} \end{aligned}$$

We can use labeled transition rule Quality\_Rec1 for process to get:

$$\begin{aligned} & n[!(v)^c.P|Q]_{l,r}^{c:C} \\ & \parallel \prod_{i \in I, n \in \mathbb{N}, k \in \mathbb{N}, k \leq n} n_i[\&_{q_i}(b_1, b_2, \dots, b_k, \dots, b_n).P_i|Q_i]_{l_i, r_i}^{c:C_i} \\ & \parallel \prod_{j \in J, m \in \mathbb{N}, h \in \mathbb{N}, h \leq m} n_j[\&_{q_j}(b_1, b_2, \dots, b_h, \dots, b_m).P_j|Q_j]_{l_j, r_j}^{c:C_j} \\ & \xrightarrow{c!!(v)[l,r]} \\ & n[P|Q]_{l,r}^{c:C} \\ & \parallel \prod_{i \in I, n \in \mathbb{N}, k \in \mathbb{N}, k \leq n} n_i[\&_{q_i}(b_1, b_2, \dots, b'_k, \dots, b_n).P_i|Q_i]_{l_i, r_i}^{c:C_i} \\ & \parallel \prod_{j \in J, m \in \mathbb{N}, h \in \mathbb{N}, h \leq m} n_j[\&_{q_j}(b_1, b_2, \dots, b_h, \dots, b_m).P_j|Q_j]_{l_j, r_j}^{c:C_j} \end{aligned}$$

as required.

Suppose that the derivation  $M \rightsquigarrow_{l,r}^c M'$  is generated by an application of the reduction rule RS-Congr:

$$\frac{M \equiv N \quad N \rightsquigarrow_{l,r}^c N' \quad N' \equiv M'}{M \rightsquigarrow_{l,r}^c M'}$$

According to the induction hypothesis, there is  $N_1$  such that  $N \xrightarrow{c!!(v)[l,r]} N_1 \equiv N'$ . Thus, the first part of this theorem tells us that there also exists  $M'$  such that  $M \xrightarrow{c!!(v)[l,r]} M_1$  and  $M_1 \equiv N_1$ . Because of the transitivity of  $\equiv$ ,  $M \xrightarrow{c!!(v)[l,r]} M_1 \equiv M'$  is valid as required.

The reduction  $M \rightsquigarrow_{l,r}^c M'$  is derived either by rule RS-Cont or rule RS-Null are straightforward.

(2) Second, the proof is by rule induction on the derivation of  $M \rightsquigarrow M'$  which is similar to the one in (1). For rule RS-Inter,

the labeled transition rule Net\_Int can do the corresponding transition and all these subnetworks will be complicated by the combination transition rule Net\_Par3 to get the final results. The remaining reduction rules can also be found the corresponding labeled transition rules easily.

The proofs for (1) and (2) are based on Lemma1, Lemma2 and Lemma3. These auxiliary lemmas and proofs can be found in Appendix A.

## V. CASE STUDY : SMART HOME AND SMART GRID

In this section, we give two real-world case studies, **Smart Home** and **Smart Grid**, to illustrate the applicability of our calculus. We focus on using CWQ to model and analyze the scenario of smart home and also briefly discuss smart grid.

### A. Modeling Smart Home in CWQ

Nowadays, in order to achieve home automation and home networking, smart home has become one of the most popular applications of WSNs. Among the many functions provided by smart home, we mainly focus on home monitoring. The architecture of smart home is shown in Figure 2. Each home has a Center Control Unit (CCU), a controller and other devices. The CCU is used for the management of all the household devices. In addition, it also collects and analyzes the data to give a useful guide to residents. All the devices are connected to CCU through wireless, and the controller also can receive warn information from CCU via wireless even though it is not at home.

$$SHome \stackrel{df}{=} Home\_Controller \parallel Home\_Monitor$$

The entire system *SHome* consists of two parallel parts, *Home\_Controller* and *Home\_Monitor*. Here, *Home\_Controller* can communicate with CCU at anytime. It uses broadcast channel  $c_1$  and has the ability to hear from channel  $c_2$  which is in its available channel set  $\{c_1, c_2\}$ . *Home\_Monitor* monitors the real-time concentrations of smoke and gas.

$$Home\_Controller \stackrel{df}{=} n[Controller]_{l,r}^{c_1:\{c_1, c_2\}}$$

$$\begin{aligned} Home\_Monitor \stackrel{df}{=} & n_{11}[CCU]_{l_{11}, r_{11}}^{c_3:\{c_1, c_3\}} \parallel \\ & n_{21}[SMonitor]_{l_{21}, r_{21}}^{c_3:\{c_3\}} \parallel \\ & n_{22}[GMonitor]_{l_{22}, r_{22}}^{c_3:\{c_3\}} \end{aligned}$$

$$Controller \stackrel{df}{=} Contr_1 \mid Others \quad Contr_1 \stackrel{df}{=} ?(reply)^{c_1}. Contr_1$$



Fig. 2: The Architecture of Smart Home

By parallel composition inside a sensor node, for process *Controller*, the subprocess *Contr<sub>1</sub>* is used to continuously monitor the warn information from CCU via channel  $c_1$ . Note that we give each process a subprocess named *Others* to do any other non-input and non-output actions.

$$\begin{aligned}
CCU &\stackrel{df}{=} CU \mid Local \mid Clock \\
CU &\stackrel{df}{=} !\langle request, id \rangle^{c_3}. local!(request, id). timer!z. \\
&\quad \&\forall(timer? \checkmark, \&\exists(? \langle reply \rangle^{c_3}, local?reply')). \\
&\quad ((0 \triangleleft reply' = none \triangleright store(reply')) \\
&\quad \triangleleft reply = none \triangleright store(reply)). \\
&\quad ((\boxtimes c_1. !\langle warn, id \rangle. \boxtimes c_3. CU) \triangleleft \\
&\quad analyze(id) = danger(id) \triangleright CU) \\
Local &\stackrel{df}{=} local?(request, id). local!avg(id, v). Local \\
Clock &\stackrel{df}{=} timer?z'. count(z'). timer! \checkmark. Clock
\end{aligned}$$

Process *CCU* also has three subprocesses. In order to monitor the concentrations of smoke and gas at home, subprocess *CU* will broadcast monitoring request to all devices periodically, while only the device that has the same *id* as the one requested for will reply its concentration value back. Due to unreliable links in wireless networks, sometimes the devices may not give a response back successfully. In order to avoid the happening of disaster accident and deadlock, we use the internal process *Local* to give a default value, which is computed from the average of its historical values. Subprocess *Clock* plays a role as a counter. Besides, *store*, *avg*, *count*, *analyze* and *danger* are functions which are intuitively defined.

$$\begin{aligned}
SMonitor &\stackrel{df}{=} SM_1 \mid Others \\
SM_1 &\stackrel{df}{=} ?(request, id)^{c_3}. (!\langle reply \rangle^{c_3}. SM_1 \triangleleft id = SM\_ID \triangleright SM_1) \\
GMonitor &\stackrel{df}{=} GM_1 \mid Others \\
GM_1 &\stackrel{df}{=} ?(request, id)^{c_3}. (!\langle reply \rangle^{c_3}. GM_1 \triangleleft id = GM\_ID \triangleright GM_1)
\end{aligned}$$

*SMonitor* stands for smoking monitor and *GMonitor* for gas monitor. They will reply the current concentration value back to CCU when they check that its id is the requested one. Here, we can see that CWQ can give a better description of smart home and make it easy to model the scenarios, making sure that CCU will not go into a deadlock state even though the reply cannot be obtained due to unreliable links.

### B. Analysis of Smart Home

In smart home, all the devices connect to CCU via wireless. Due to unreliable links in wireless networks, devices may not send critical data back successfully, which makes CCU go into a deadlock state or makes the home monitoring go into a very dangerous situation. Based on the model given in A, in this subsection, we use operational semantics to analyze the deadlock free property by giving default values from the local process.

Figure 3 shows a dangerous situation. In some period, CCU broadcasts a request about collecting the current concentration of smoke. However, due to unreliable links in wireless networks, smoke monitor cannot send data back. To avoid CCU going into a deadlock state or missing some important safety-critical value, after timeout, the local process in CCU will send a default value of the smoke concentration, which is computed according to the average recent smoke concentration data to CCU. Then the CCU may not go into a deadlock state

or a dangerous situation because of lacking monitoring data. The entire model of home monitor is given below:

$$\begin{aligned}
HMonitor &\stackrel{df}{=} n_{11}[CCU]_{l_{11}, r_{11}}^{c_3: \{c_1, c_3\}} \parallel n_{21}[SMonitor]_{l_{21}, r_{21}}^{c_3: \{c_3\}} \\
&\quad \parallel n_{22}[GMonitor]_{l_{22}, r_{22}}^{c_3: \{c_3\}} \\
&\stackrel{df}{=} n_{11}[CU|Local|Clock]_{l_{11}, r_{11}}^{c_3: \{c_1, c_3\}} \parallel \\
&\quad n_{21}[SM_1|Others]_{l_{21}, r_{21}}^{c_3: \{c_3\}} \parallel n_{22}[GM_1|Others]_{l_{22}, r_{22}}^{c_3: \{c_3\}} \\
&\stackrel{df}{=} n_{11}[\langle request, n_{21} \rangle^{c_3}. local!(request, n_{21}). timer!y. \\
&\quad \&\forall(timer? \checkmark, \&\exists(? \langle reply \rangle^{c_3}, local?reply')) \\
&\quad ((0 \triangleleft reply' = none \triangleright store(reply')) \\
&\quad \triangleleft reply = none \triangleright store(reply)). \\
&\quad (\boxtimes c_1. !\langle warn, n_{21} \rangle. \boxtimes c_3. CU) \triangleleft \\
&\quad analyze(n_{21}) = Danger(n_{21}) \triangleright CU \mid \\
&\quad local?(request, n_{21}). local!avg(n_{21}, v). Local \mid \\
&\quad timer?y'. count(y'). timer! \checkmark. Clock]_{l_{11}, r_{11}}^{c_3: \{c_1, c_3\}} \parallel \\
&\quad n_{21}[\langle request, n_{21} \rangle^{c_3}. (!\langle reply \rangle^{c_3}. SM_1) \triangleleft \\
&\quad n_{21} = n_{21} \triangleright SM_1 \mid Others]_{l_{21}, r_{21}}^{c_3: \{c_3\}} \parallel \\
&\quad n_{22}[\langle request, n_{21} \rangle^{c_3}. (!\langle reply \rangle^{c_3}. GM_1) \triangleleft \\
&\quad n_{21} = n_{22} \triangleright GM_1 \mid Others]_{l_{22}, r_{22}}^{c_3: \{c_3\}}
\end{aligned}$$

We give one possible labeled transition situation. Here, we only give the communications between nodes because the internal events inside one node cannot be seen from network perspective directly, which is shown as a  $\tau$  action. The process of analysis is as follows.

$$\begin{aligned}
HMonitor &\xrightarrow{c_3! \langle request, n_{21} \rangle [l_{11}, r_{11}]} \\
&\quad n_{11}[local!(request, n_{21}). timer!y. \\
&\quad \&\forall(timer? \checkmark, \&\exists(? \langle reply \rangle^{c_3}, local?reply')) \\
&\quad ((0 \triangleleft reply' = none \triangleright store(reply')) \\
&\quad \triangleleft reply = none \triangleright store(reply)). \\
&\quad (\boxtimes c_1. !\langle warn, n_{21} \rangle. \boxtimes c_3. CU) \triangleleft \\
&\quad analyze(n_{21}) = Danger(n_{21}) \triangleright CU \mid \\
&\quad local?(request, n_{21}). local!avg(n_{21}, v). Local \\
&\quad timer?y'. count(y'). timer! \checkmark. Clock]_{l_{11}, r_{11}}^{c_3: \{c_1, c_3\}} \parallel \\
&\quad n_{21}[\langle request, n_{21} \rangle^{c_3}. SM_1) \triangleleft n_{21} = n_{21} \triangleright SM_1 \mid Others]_{l_{21}, r_{21}}^{c_3: \{c_3\}} \parallel \\
&\quad n_{22}[\langle request, n_{21} \rangle^{c_3}. GM_1) \triangleleft n_{21} = n_{22} \triangleright GM_1 \mid Others]_{l_{22}, r_{22}}^{c_3: \{c_3\}} \\
&\stackrel{df}{=} HMonitor_1
\end{aligned}$$

Then, the sender will do some internal actions, e.g., sending messages to local process and clock. After these actions, the process will be evolved into the subprocess with quality predicate. Due to unreliable links in wireless networks, CCU may not receive data from smoke monitor successfully, that is:

$$\begin{aligned}
HMonitor_1 &\xrightarrow{\tau, \tau} c_3! \langle reply \rangle [l_{21}, r_{21}] \\
&\quad n_{11}[\&\forall(timer? \checkmark, \&\exists(? \langle reply \rangle^{c_3}, local?reply')) \\
&\quad ((0 \triangleleft reply' = none \triangleright store(reply')) \\
&\quad \triangleleft reply = none \triangleright store(reply)).
\end{aligned}$$

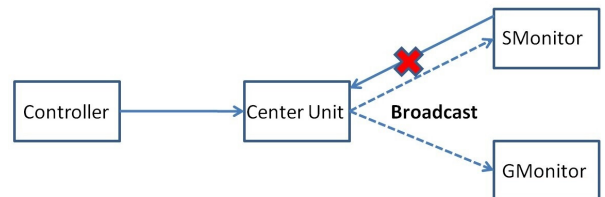


Fig. 3: One Dangerous Monitoring Situation

$$\begin{aligned}
& (\bowtie c_1. !\langle warn, n_{21} \rangle. \bowtie c_3. CU) \triangleleft \\
& analyze(n_{21}) = Danger(n_{21}) \triangleright CU \mid \\
& local!avg(n_{21}, v). Local \mid \\
& count(y'). timer!\checkmark. Clock]_{l_{11}, r_{11}}^{c_3: \{c_1, c_3\}} \parallel \\
& n_{21}[SM_1 \mid Others]_{l_{21}, r_{21}}^{c_3: \{c_3\}} \parallel \\
& n_{22}[GM_1 \mid Others]_{l_{22}, r_{22}}^{c_3: \{c_3\}} \stackrel{df}{=} HMonitor_2
\end{aligned}$$

In order to make sure the system does not go into a deadlock state or missing some safety-critical data because of node failing and to increase the service quality offered by the system, after timeout, local process in CCU will send a default value that is computed according to the average of recent data of smoke monitor to CCU. When CCU receives the value from local process, it will do some computation to finish the whole work.

$$HMonitor_2 \xrightarrow{\tau}^* HMonitor$$

Hence, CWQ gives a better modeling and analysis of smart home, and avoid the system deadlock due to unreliable links in wireless networks. Even though the default value is not as good as the fresh one, but it still improves the quality of the whole system.

### C. Discussion on Smart Grid

Smart Grid is a large-scale outdoor case study in real life. It is a new-style power system of smart city and has already been constructed in several high-tech industrial zones. It monitors electric transmission line as well as providing a guide for intelligent use of electricity. The architecture of smart grid is shown in figure 4.

Collecting and analyzing the power consumption data is one of the most important functions of smart grid. Taking one residence building into consideration, the center electricity data collector will collect the consumption data of the entire building every 30 minutes. However, data lacking is very common due to unreliable links in wireless networks. High rate of lacking data will make the collector to give a wrong guide or even not give a guide. Thus, it is critical to avoid data lacking. We assume there are  $n$  people living in a building and use CWQ to formalize the collector part below.

$$Collector \stackrel{df}{=} \prod_{i \in \{0..n\}, n \in \mathbb{N}} !\langle request_i, id_i \rangle^{c_3}. local_i!(request_i, id_i).$$

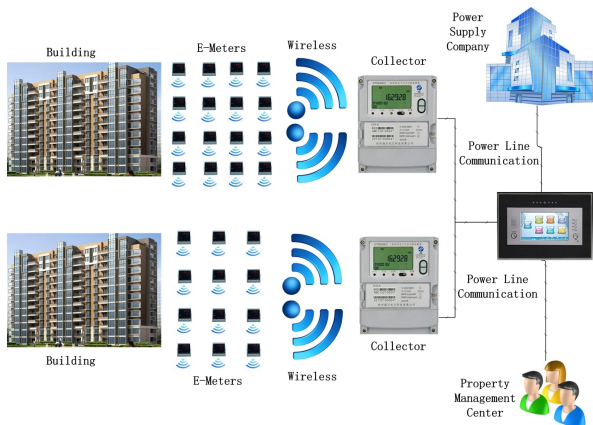


Fig. 4: The Architecture of Smart Grid

$$\begin{aligned}
& timer_i!x_i. \& \forall (timer_i? \checkmark, \& \exists (? (reply_i)^{c_3}, local_i?reply'_i)). \\
& ((0 \triangleleft reply'_i = none \triangleright store(y_i)) \\
& \triangleleft reply_i = none) \triangleright (store'(y_i)) \mid \\
& \prod_{i \in \{0..n\}, n \in \mathbb{N}} local_i?(request_i, id_i). local_i!avg(id_i, v_i) \mid \\
& \prod_{i \in \{0..n\}, n \in \mathbb{N}} timer_i?x_i. count(x_i). timer_i!\checkmark
\end{aligned}$$

The modeling and analysis for smart grid is similar to that for smart home discussed in subsections A and B, thus we omit the details here. Through the default value mechanism in CWQ, data lacking is avoided which gives users a better guide of intelligent use of electricity. These two case studies illustrate the applicability of our calculus for both indoor scenario and outdoor scenario.

## VI. CONCLUSION AND FUTURE WORK

In this paper, we proposed a calculus for WSNs from a quality perspective. We combined local broadcast with quality predicate to ensure that every node in the network can behave in a reasonable manner even though they are in an unreliable communication. Default values are given in the case that ideal behaviors of sensor nodes fail to increase the service quality offered by the whole system. Moreover, we gave the operational semantics of CWQ in terms of both LTS and RS, and proved a correspondence result between them. Finally, examples and two real world case studies about smart home and smart grid were given to illustrate the applicability of this calculus.

Due to different default value will bring different performance, in the future, we are going to explore how to choose a better default value to give a high quality performance of the system. Moreover, we are also going to combine the mobility together with our calculus using differential equation. To assist in analyzing properties, tool supporting will also be developed.

## ACKNOWLEDGMENT

This work was partly supported by the Danish National Research Foundation and the National Natural Science Foundation of China (Grant No. 61361136002) for the Danish-Chinese Center for Cyber Physical Systems. It was also supported by National High Technology Research and Development Program of China (Grant No. 2012AA011205), National Natural Science Foundation of China (Grant No. 61321064), Shanghai Knowledge Service Platform Project (No. ZF1213) and Shanghai Minhang Talent Project.

## REFERENCES

- [1] S. D. Brookes, C. A. R. Hoare, and A. W. Roscoe. A Theory of Communicating Sequential Processes. *J. ACM*, 31(3):560–599, 1984.
- [2] A. Fehnker, R. J. van Glabbeek, P. Höfner, A. McIver, M. Portmann, and W. L. Tan. A Process Algebra for Wireless Mesh Networks. In *ESOP*, volume 7211 of *LNCS*, pages 295–315. Springer, 2012.
- [3] J. C. Godskesen. A Calculus for Mobile Ad Hoc Networks. In *COORDINATION*, volume 4467 of *LNCS*, pages 132–150. Springer, 2007.
- [4] C. A. R. Hoare. *Communicating Sequential Processes*. Prentice-Hall, 1985.
- [5] I. Lanese and D. Sangiorgi. An operational semantics for a calculus for wireless systems. *Theor. Comput. Sci.*, 411(19):1928–1948, 2010.

- [6] S. Liu, X. Wu, Q. Li, H. Zhu, and Q. Wang. Formal Approaches to Wireless Sensor Networks. In *SSIRI*, pages 11–18. IEEE, 2011.
- [7] M. Merro. An Observational Theory for Mobile Ad Hoc Networks (full version). *Inf. Comput.*, 207(2):194–208, 2009.
- [8] N. Mezzetti and D. Sangiorgi. Towards A Calculus for Wireless Systems. *Electr. Notes Theor. Comput. Sci.*, 158:331–353, 2006.
- [9] R. Milner. *A Calculus of Communicating Systems*, volume 92 of *LNCS*. Springer, 1980.
- [10] S. Nanz and C. Hankin. A Framework for Security Analysis of Mobile Wireless Networks. *Theor. Comput. Sci.*, 367(1-2):203–227, 2006.
- [11] H. R. Nielson, F. Nielson, and R. Vigo. A Calculus for Quality. In *FACS*, volume 7684 of *LNCS*, pages 188–204. Springer, 2012.
- [12] K. Ostrovsky, K. V. S. Prasad, and W. Taha. Towards A Primitive Higher Order Calculus of Broadcasting Systems. In *PPDP*, pages 2–13. ACM, 2002.
- [13] K. V. S. Prasad. A Calculus of Broadcasting Systems. *Sci. Comput. Program.*, 25(2-3):285–327, 1995.
- [14] R. Vigo, F. Nielson, and H. R. Nielson. Broadcast, Denial-of-Service, and Secure Communication. In *IFM*, volume 7940 of *LNCS*, pages 412–427. Springer, 2013.

#### APPENDIX A: AUXILIARY LEMMA AND PROOF

**Lemma 1.** If  $M \xrightarrow{c??(v)@l} M'$ , then

$$M \equiv \prod_{i \in I, k \in \mathbb{N}, n \in \mathbb{N}, k \leq n} n_i[\&_{q_i}(b_1, b_2, \dots, b_k, \dots, b_n).P_i|Q_i]_{l_i, r_i}^{c:C_i} \parallel \prod_{j \in J, h \in \mathbb{N}, m \in \mathbb{N}, h \leq m} n_j[\&_{q_j}(b_1, b_2, \dots, b_h, \dots, b_m).P_j|Q_j]_{l_j, r_j}^{c':C_j} \parallel M''$$

Where  $\forall i \in I. D(l, l_i) \wedge c \in C_i \leq r$ ,  $\forall j \in J. D(l, l_j) > r \vee c \notin C_j$  and  $(l, r, c) \not\models M''$ . Furthermore

$$M' \equiv \prod_{i \in I, k \in \mathbb{N}, n \in \mathbb{N}, k \leq n} n_i[\&_{q_i}(b_1, b_2, \dots, b'_k, \dots, b_n).P_i|Q_i]_{l_i, r_i}^{c:C_i} \parallel \prod_{j \in J, h \in \mathbb{N}, m \in \mathbb{N}, h \leq m} n_j[\&_{q_j}(b_1, b_2, \dots, b_h, \dots, b_m).P_j|Q_j]_{l_j, r_j}^{c':C_j} \parallel M''$$

**Proof.** The proof is based on the induction of the derivation  $M \xrightarrow{c??(v)@l} M'$ .

For rule *Net\_Null1*, Lemma 1 is obviously valid. Here,  $I = J = \phi$  and  $M'' = 0$ .

For rule *Net\_Rcv1*, we can get  $P \xrightarrow{?(v)^c} P'$  from promise, which involves in three process transition rules in Table 3, *Bro\_Receive*, *Qli\_Bro1* and *Qli\_Bro2*. Here,  $I = \{i\}$ ,  $J = \phi$  and  $N'' = 0$ .

From the premise of rule *Net\_Rcv2*, we know that the node either satisfies  $D(l, l') > r'$  or  $c \notin C'$ . If the node satisfies the former case, we can get  $I = \phi$ ,  $J = \{j\}$  and  $M'' = 0$ , otherwise, we can put the node into the network  $M''$ .

Rule *Net\_Rcv3* is an inductive rule, which combines the corresponding index sets subnetworks together with the non-index part of the network  $M''$  in the previous premises into the desired form.

**Lemma 2.** If  $M \xrightarrow{c!!(v)[l, r]} M'$ , then

$$M \equiv n[!(v)^c.P|Q]_{l, r}^{c:C} \parallel \prod_{i \in I, k \in \mathbb{N}, n \in \mathbb{N}, k \leq n} n_i[\&_{q_i}(b_1, b_2, \dots, b_k, \dots, b_n).P_i|Q_i]_{l_i, r_i}^{c:C_i} \parallel \prod_{j \in J, h \in \mathbb{N}, m \in \mathbb{N}, h \leq m} n_j[\&_{q_j}(b_1, b_2, \dots, b_h, \dots, b_m).P_j|Q_j]_{l_j, r_j}^{c':C_j} \parallel M''$$

Where  $\forall i \in I. D(l, l_i) \leq r \wedge c \in C_i$ ,  $\forall j \in J. D(l, l_j) > r \vee c \notin C_j$  and  $(l, r, c) \not\models M''$ . Furthermore

$$M' \equiv n[P|Q]_{l, r}^{c:C} \parallel \prod_{i \in I, k \in \mathbb{N}, n \in \mathbb{N}, k \leq n} n_i[\&_{q_i}(b_1, b_2, \dots, b'_k, \dots, b_n).P_i|Q_i]_{l_i, r_i}^{c:C_i} \parallel \prod_{j \in J, h \in \mathbb{N}, m \in \mathbb{N}, h \leq m} n_j[\&_{q_j}(b_1, b_2, \dots, b_h, \dots, b_m).P_j|Q_j]_{l_j, r_j}^{c':C_j} \parallel M''$$

**Proof.** The proof is similar to the one for Lemma 1, and we use Lemma 1 to deal with the input transition premises.

**Lemma 3.** If  $M \xrightarrow{\tau} M'$ , then

$$M \equiv \prod_{i \in I, k \in \mathbb{N}, n \in \mathbb{N}, k \leq n} n_i[\&_{q_i}(b_1, b_2, \dots, b_k, \dots, b_n).P_i|t_k!v_k.Q_i]_{l_i, r_i}^{c_i:C_i} \parallel \prod_{j, j' \in J} n_j[P_j \triangleleft u = y \triangleright P_{j'}|Q_j]_{l_j, r_j}^{c_j:C_j} \parallel \prod_{s \in S} n_s[0]_{l_s, r_s}^{c_s:C_s}$$

Where  $\forall i \in I, \forall j \in J, \forall s \in S$  and  $u = y$ . Furthermore

$$M' \equiv \prod_{i \in I, k \in \mathbb{N}, n \in \mathbb{N}, k \leq n} n_i[\&_{q_i}(b_1, b_2, \dots, b'_k, \dots, b_n).P_i|Q_i]_{l_i, r_i}^{c_i:C_i} \parallel \prod_{j, j' \in J} n_j[P_j|Q_j]_{l_j, r_j}^{c_j:C_j} \parallel \prod_{s \in S} n_s[0]_{l_s, r_s}^{c_s:C_s}$$

**Proof.** The proof is by induction on the size of  $I \cup J \cup S$  and for each case it is similar to the one in Lemma 1, using rules for internal actions.